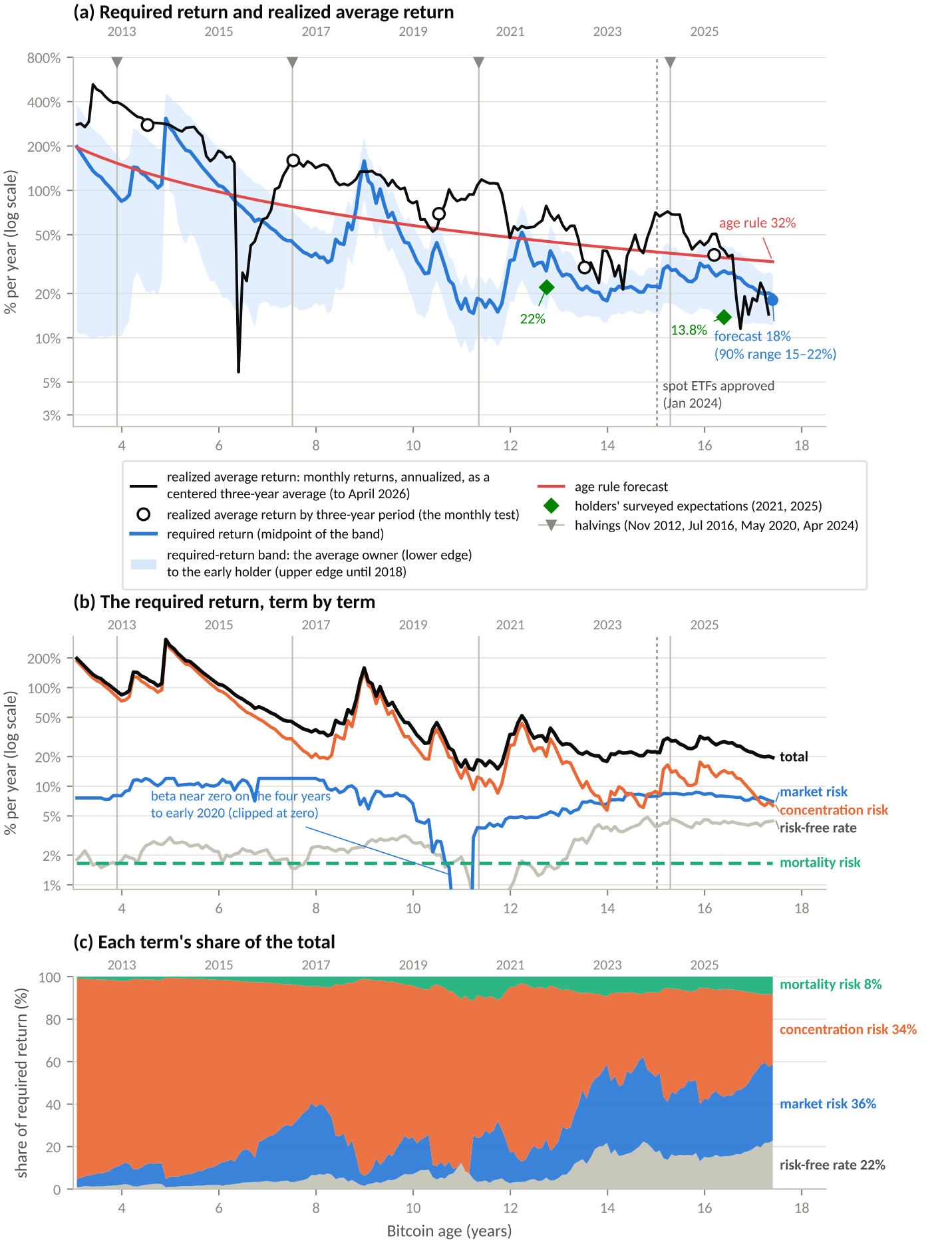

(a) Required return and realized average return
2013
2015
2017
2019
2021
2023
2025
800%
400%
200%
100%
50%
20%
10%
5%
3%
% per year (log scale)
age rule 32%
22%
13.8%
forecast 18%
(90% range 15–22%)
spot ETFs approved
(Jan 2024)
4
6
8
10
12
14
16
18
realized average return: monthly returns, annualized, as a centered three-year average (to April 2026)
realized average return by three-year period (the monthly test)
required return (midpoint of the band)
required-return band: the average owner (lower edge) to the early holder (upper edge until 2018)
age rule forecast
holders' surveyed expectations (2021, 2025)
halvings (Nov 2012, Jul 2016, May 2020, Apr 2024)
(b) The required return, term by term
200%
100%
50%
20%
10%
5%
2%
1%
beta near zero on the four years to early 2020 (clipped at zero)
total
market risk
concentration risk
risk-free rate
mortality risk
(c) Each term's share of the total
100
80
60
40
20
0
share of required return (%)
mortality risk 8%
concentration risk 34%
market risk 36%
risk-free rate 22%
Bitcoin age (years)